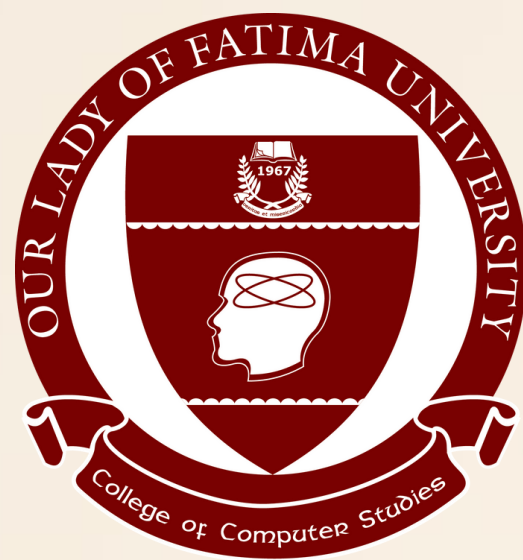




Module 10

Introduction to Cybersecurity

ITCL112 - INTRODUCTION TO COMPUTING W/ LAB

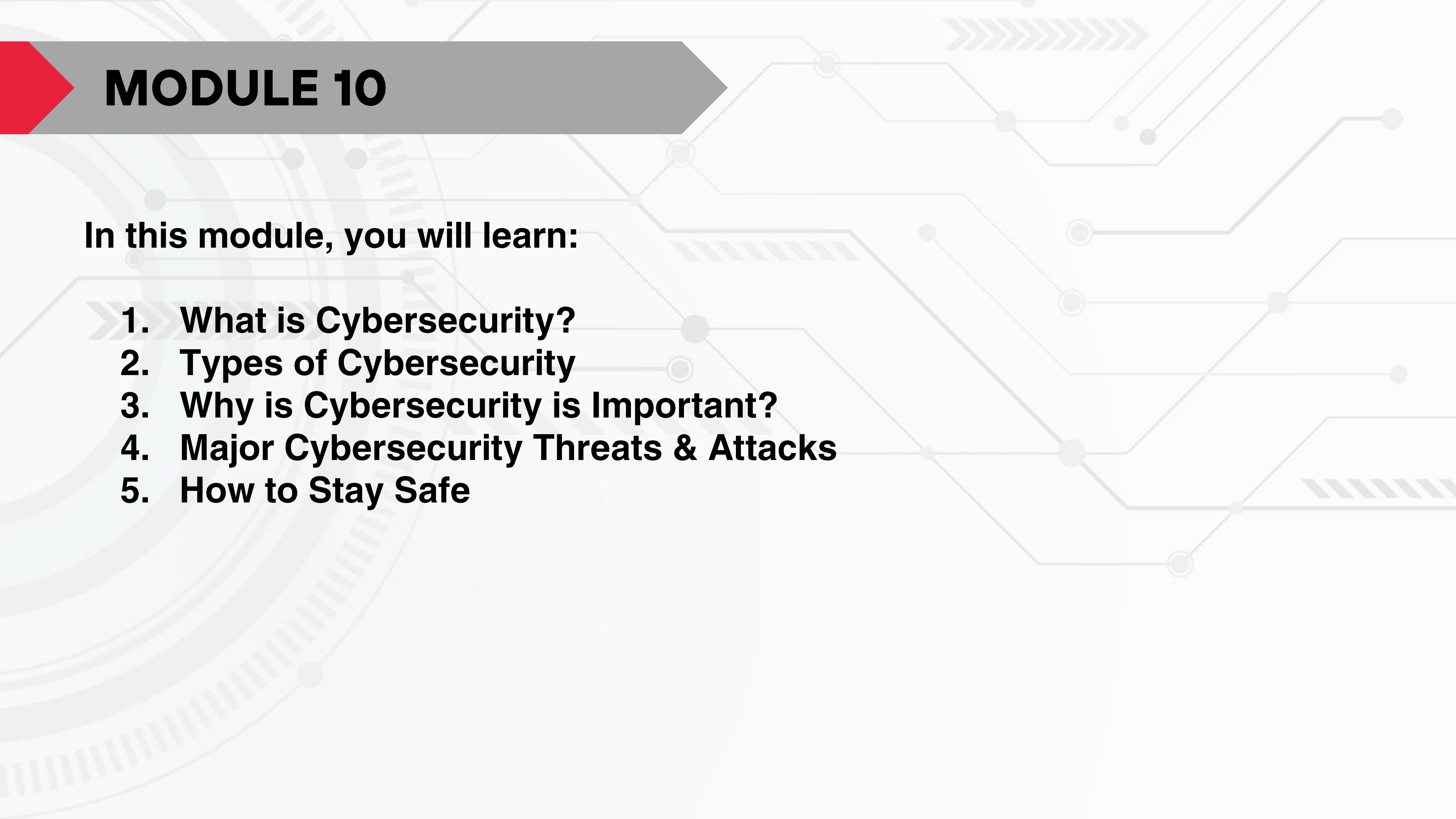
INTRODUCTION



Cybersecurity is about keeping computers, networks, and information safe from online threats like viruses, hackers, and scams. It protects data from being stolen or damaged and makes sure systems work properly. In today's digital world, cybersecurity is important for protecting personal information, school or business files, and even government systems.

MODULE 10

In this module, you will learn:

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- 1. » What is Cybersecurity?**
 - 2. Types of Cybersecurity**
 - 3. Why is Cybersecurity is Important?**
 - 4. Major Cybersecurity Threats & Attacks**
 - 5. How to Stay Safe**

What is Cyber Security?



Cybersecurity is the practice of protecting digital devices, networks, and sensitive data from cyber threats such as hacking, malware, and phishing attacks." It involves a range of strategies, technologies, and best practices designed to safeguard computers, networks, and data from cyber attacks.

Cybersecurity involves using specialized tools to detect and remove harmful software while also learning to identify and avoid online scams. Practicing good cybersecurity habits helps keep your data private and ensures a safe online experience. It's also referred to as Information Security (INFOSEC), Information Assurance (IA), or System Security.

Types of Cybersecurity



1. Network Security

It focuses on securing computer networks from unauthorized access, data breaches, and other network-based threats. This involves implementing technologies such as Firewalls, Intrusion detection systems (IDS), Virtual private networks (VPNs), and Network segmentation as well as deploying antivirus software

- Using public Wi-Fi in locations like cafes and malls poses significant security risks. Malicious actors on the same network can potentially intercept your online activity, including sensitive information. If you use payment gateways on these unsecured networks, your financial data could be compromised because these open networks don't have proper security layers, which means anyone even hackers can watch what you're doing online.
- So, use a secure private network or VPN to protect your internal network from outside threats

Types of Cybersecurity



2. Application Security

Concerned with securing software applications and preventing vulnerabilities that could be exploited by attackers. It involves secure coding practices, regular software updates and patches, and application-level firewalls.

- Most of the Apps that we use on our cell phones are Secured and work under the rules and regulations of the Google Play Store.
- There are 3.553 million applications in Google Play, Apple App Store has 1.642 million, and Amazon App Store has 483 million available for users to download. With so many choices, it's easy to assume all apps are safe but that's not true.
- Some apps pretend to be secure, but once installed, they collect personal data and secretly share it with third-party companies.
- The app must be installed from a trustworthy platform, not from some 3rd party website in the form of an APK (Android Application Package).

Types of Cybersecurity



3. Information or Data Security

Focuses on protecting sensitive information from unauthorized access, disclosure, alteration, or destruction. It includes Encryption, Access controls, Data classification, and Data loss prevention (DLP) measures.

- Incident response refers to the process of detecting, analyzing, and responding to security incidents promptly.
- Promoting security awareness among users is essential for maintaining information security. It involves educating individuals about common security risks, best practices for handling sensitive information, and how to identify and respond to potential threats like phishing attacks or social engineering attempts.
- Encryption is the process of converting information into an unreadable format (ciphertext) to protect it from unauthorized access.

Types of Cybersecurity



4. Cloud Security

It involves securing data, applications, and infrastructure hosted on cloud platforms, and ensuring appropriate access controls, data protection, and compliance. It uses various cloud service providers such as AWS, Azure, Google Cloud, etc., to ensure security against multiple threats.

- Cloud-based data storage has become a popular option over the last decade. It enhances privacy if configured and managed correctly and saves data on the cloud, making it accessible from any device with proper authentication.
- These platforms offer free tiers for limited usage, and users must pay for additional storage or services
- It is a cloud service provider that offers a wide range of services, including storage, computing, and security tools.

Types of Cybersecurity



5. Endpoint Security

Refers to securing individual devices such as computers, laptops, smartphones, and IoT devices. It includes antivirus software, intrusion prevention systems (IPS), device encryption, and regular software updates.

- Antivirus and Anti-malware software that scans and detects malicious software, such as Viruses, Worms, Trojans, and Ransomware. These tools identify and eliminate or quarantine malicious files, protecting the endpoint and the network from potential harm.
- Firewalls are essential components of endpoint security. They monitor and control incoming and outgoing network traffic, filtering out potentially malicious data packets.
- Keeping software and operating systems up to date with the latest security patches and updates is crucial for endpoint security.

Types of Cybersecurity



6. Operational Security

Refers to the processes and policies organizations implement to protect sensitive data from internal threats and human errors. It involves access controls, risk management, employee training, and monitoring activities to prevent data leaks and security breaches.

- Access Controls ensure that only authorized personnel can access critical systems and sensitive information. This includes role-based access, multi-factor authentication (MFA), and least privilege principles.
- Risk Management involves identifying, analyzing, and mitigating security risks within an organization. It includes regular security assessments, vulnerability testing, and compliance audits.
- Employee Training is crucial for preventing insider threats and social engineering attacks. Organizations conduct cybersecurity awareness programs to educate employees on phishing scams, password security, and data handling best practices.
- Monitoring & Incident Response includes tracking user activity, detecting suspicious behavior, and responding to security incidents in real time. Security Information and Event Management (SIEM) tools help organizations analyze and mitigate threats effectively.

Types of Cybersecurity



7. Internet of Things (IoT) Security

Refers to protecting internet-connected devices such as smart home gadgets, industrial sensors, medical equipment, and wearable technology from cyber threats. IoT security ensures that these devices do not become entry points for hackers to exploit networks and steal sensitive data.

- Device Authentication & Encryption ensures that only authorized devices can connect to networks. Encryption protects data transmitted between IoT devices and servers from interception.
- Firmware & Software Updates are crucial to patch security vulnerabilities. Regular updates help prevent exploitation by cybercriminals who target outdated IoT firmware.
- Network Segmentation isolates IoT devices from critical systems, reducing the risk of widespread attacks if one device is compromised. This approach limits unauthorized access and lateral movement within a network.
- IoT Security Standards & Compliance include implementing industry security frameworks like Zero Trust Architecture (ZTA) and following best practices such as strong password policies, secure APIs, and endpoint protection to enhance IoT device security.

Why is Cybersecurity is Important?



IMPORTANT

Cyber Security is important because the government, corporations, and medical organizations, collect military, financial, process, and store unprecedented amounts of data on a computer and other properties like personal information, and this private information exposure could have negative consequences.

In 1972, when the internet was just starting (called ARPANET at the time), a test virus named Creeper was created and then another program called Reaper was made to remove it. This early experiment showed why digital security was needed and helped start what we now call cybersecurity.

Why is Cybersecurity is Important?



IMPORTANT

Rising Cyber Threats: How Hackers Exploit Weak Security

Cybercriminals constantly look for ways to exploit weaknesses in systems, networks, and personal devices. Weak passwords, outdated software, and unsecured networks give them easy access. They use advanced tactics like phishing, ransomware, and social engineering to steal data, disrupt systems, or demand money.

With AI-driven threats, even bots can breach security, making cybersecurity more important than ever.

Example: While shopping online, we often save details like email, address, and card info for convenience. Later, we might receive a fake email claiming a special offer from a well-known site like Flipkart or Amazon. When we enter our saved details to claim the offer, hackers can steal that data and misuse it, causing financial loss.

Why is Cybersecurity is Important?



IMPORTANT

Consequences of Cyber Attacks for Businesses & Individuals

A successful cyber attack can have devastating effects, both financially and reputationally. For businesses, a data breach can lead to massive financial losses, legal penalties, and loss of customer trust. Small businesses are especially vulnerable, as they often lack robust security measures. Individuals, on the other hand, face risks like identity theft, financial fraud, and personal data leaks.

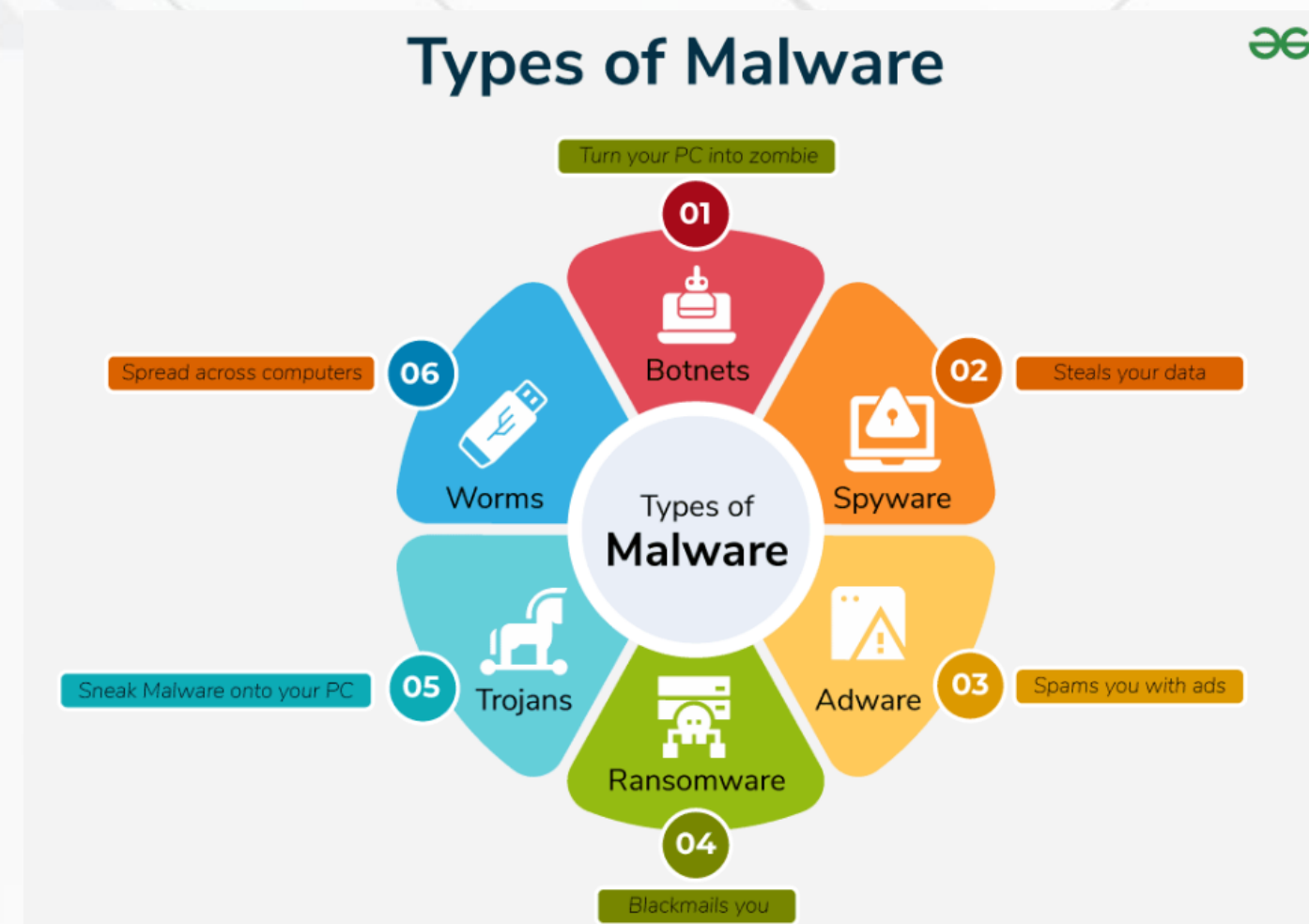
Cyber attacks can wipe out bank accounts, expose private information, and even lock users out of their own devices unless a ransom is paid. The consequences can be long-lasting, leading to emotional distress and financial instability.

Major Cybersecurity Threats & Attacks



1. Malware Attacks

Malware is software that infects systems without user consent to steal sensitive data (bank details, passwords, personal emails), disrupt operations, or alter core system behavior. It can exfiltrate confidential information, corrupt or delete files, and impair system availability or integrity. Examples include ransomware (encrypts files for ransom) and spyware (monitors activity); mitigate risks with up-to-date antivirus and caution when opening links or attachments.

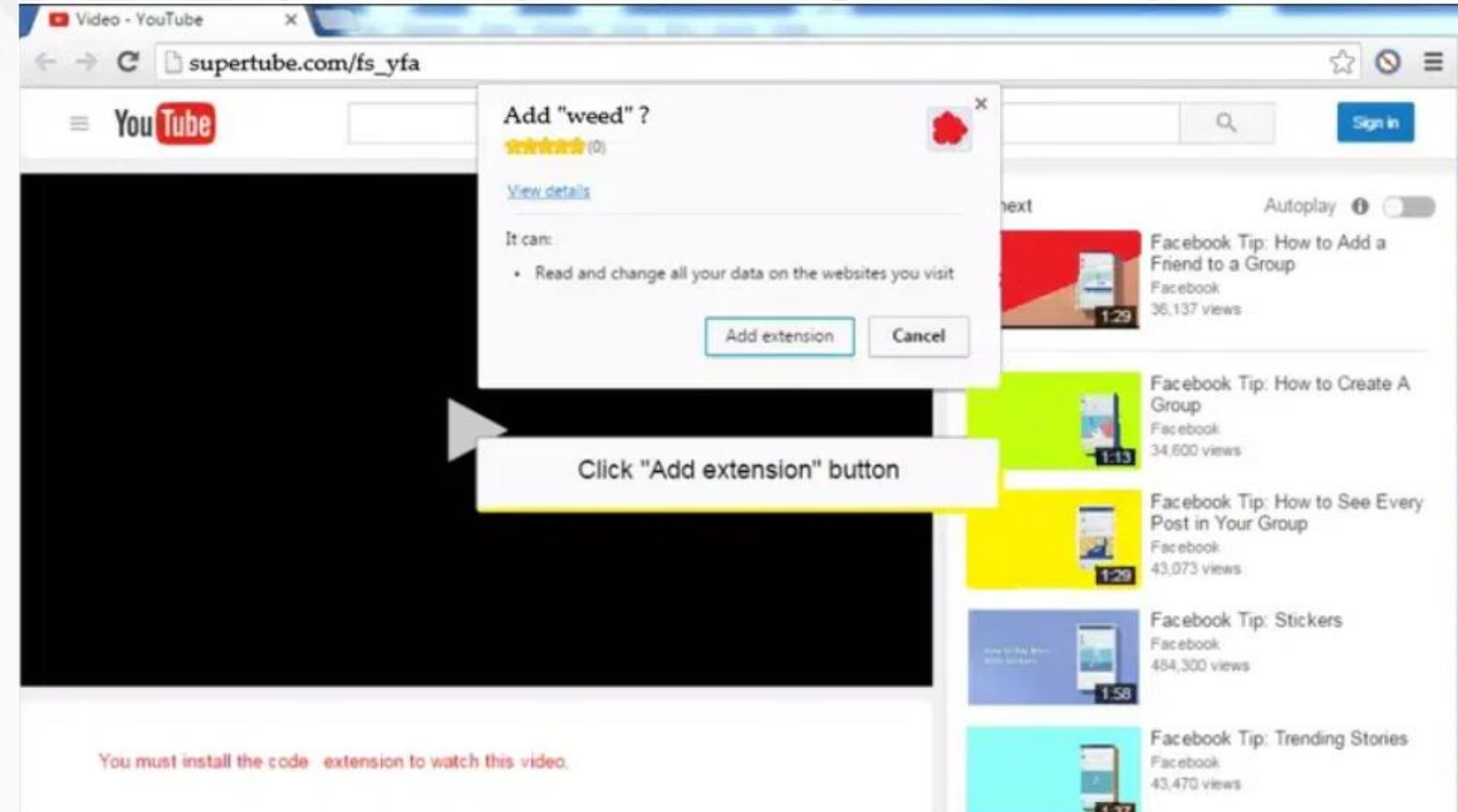


Major Cybersecurity Threats & Attacks

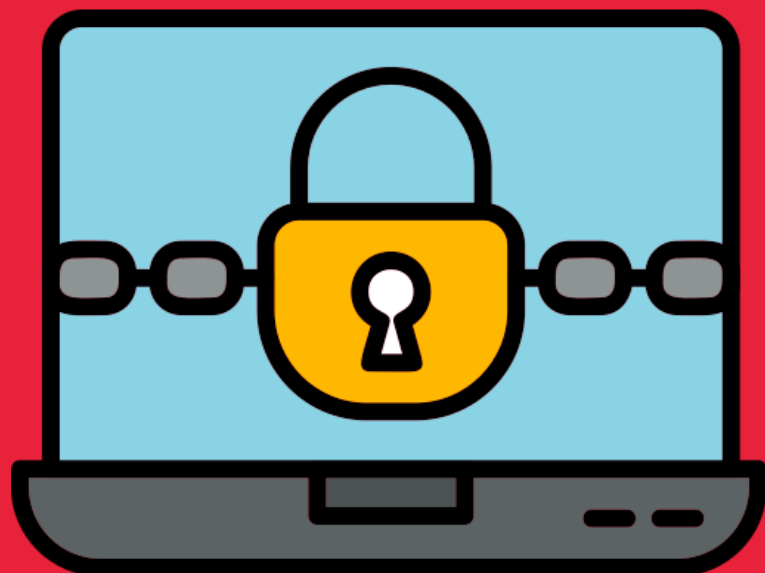


2. Phishing & Spear Phishing Attacks

Phishing is another type of cyber attack. Phishing got its name from "**phish**" meaning fish. It's a common phenomenon to put bait for the fish to get trapped. Similarly, phishing works. It is an unethical way to dupe the user or victim to click on harmful sites. The attacker crafts the harmful site in such a way that the victim feels it to be an authentic site, thus falling prey to it. The most common mode of phishing is by sending spam emails that appear to be authentic and thus, taking away all credentials from the victim.

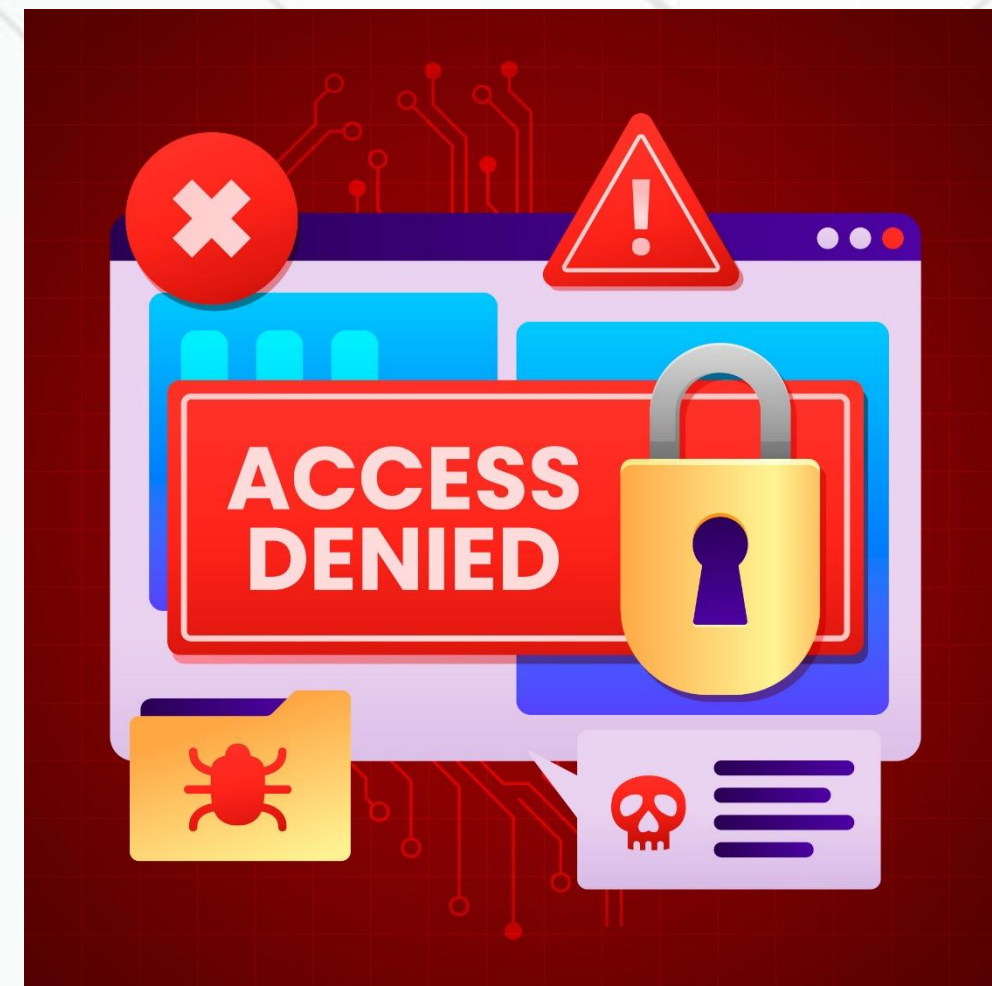


Major Cybersecurity Threats & Attacks



3. Ransomware Attacks

Ransomware is a form of malicious software that prevents computer users from accessing their data by encrypting it. Cybercriminals use it to ransom money from individuals or organizations whose data they have hacked, and they hold the data hostage until the ransom is paid. If the cybercriminals do not pay the ransom within the specified time frame, the data may leak to the public or be permanently damaged. One of the most serious issues that businesses face is ransomware.



Major Cybersecurity Threats & Attacks

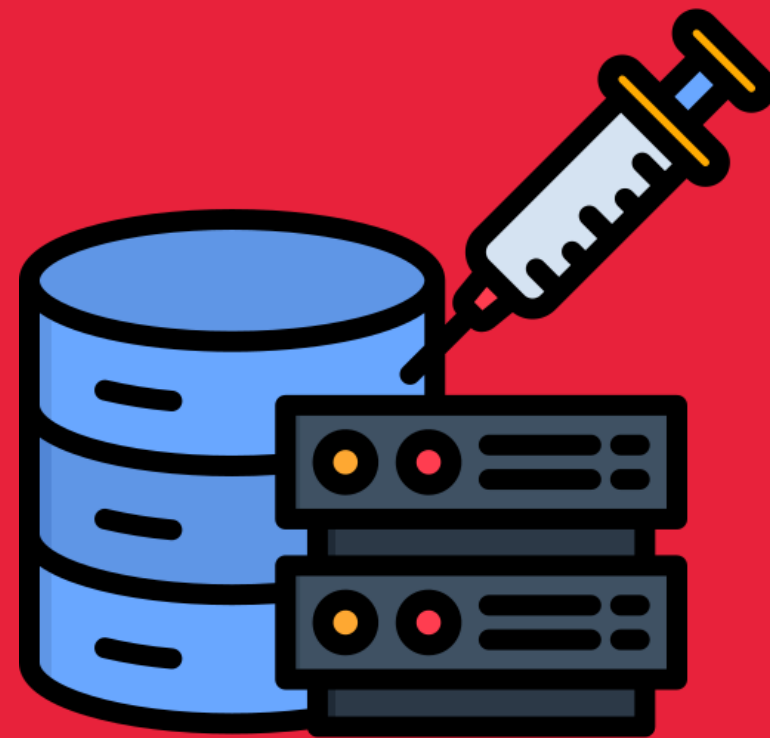


4. DoS Attacks and Ddos Attacks

Denial of service (DOS) is a network security attack, in which, the hacker makes the system or data unavailable to someone who needs it. Hacker tries to make a network, system, or machine unavailable by flooding it with fake requests or traffic. This prevents real users from accessing it, causing anything from slowdowns to complete shutdowns.

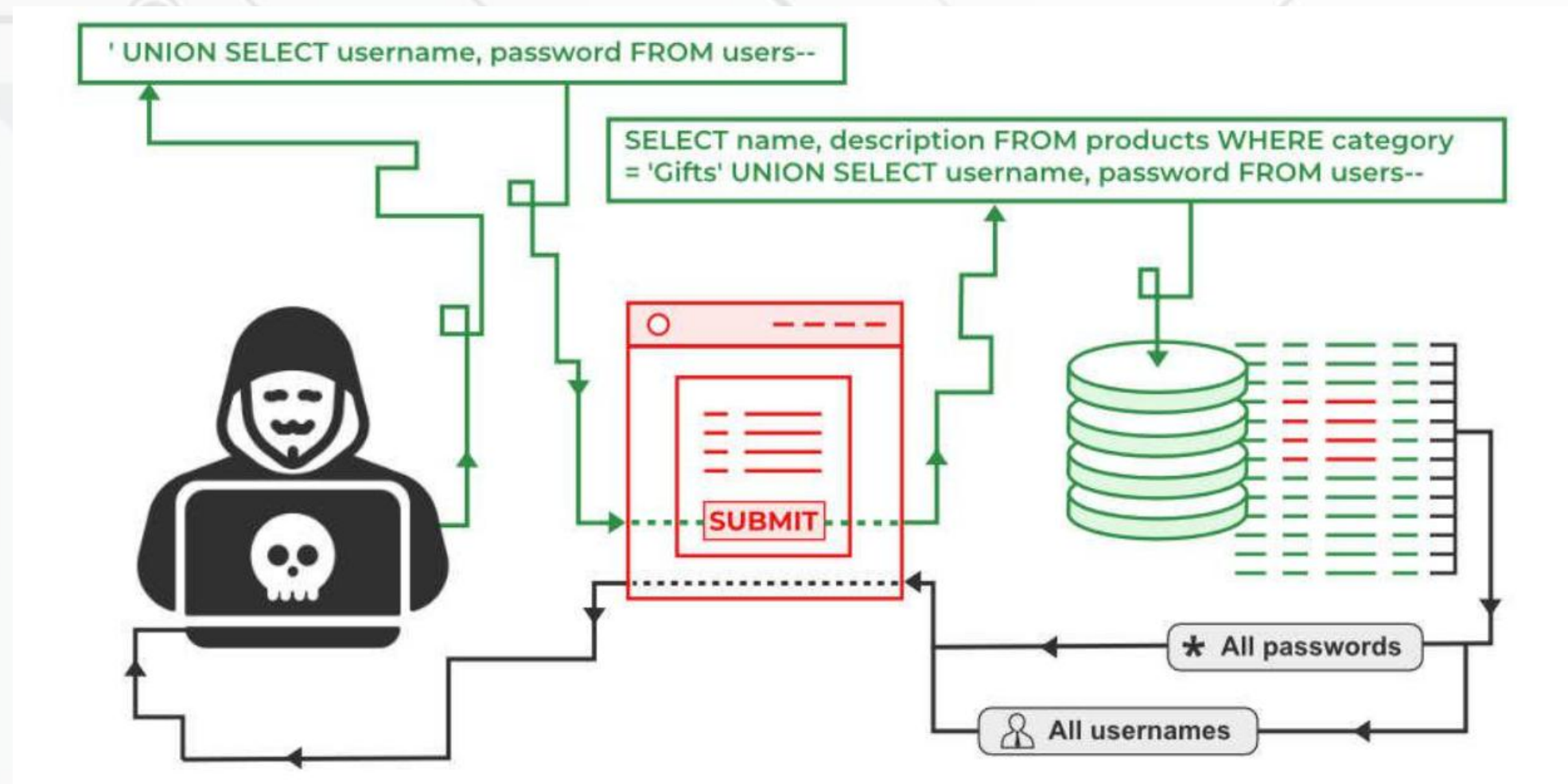
Distributed Denial-of-Service (DDoS) Attacks: DDoS attacks use multiple systems, often compromised computers (botnets), to attack a single target. Examples are amplification attacks and botnet-based attacks. In an amplification attack, attackers use services like DNS to send a small query that generates a large response, flooding the victim with data. Botnets coordinate many infected computers to send attack traffic from multiple sources, making it hard to defend against.

Major Cybersecurity Threats & Attacks



5. SQL Injection (SQLi) & NoSQL Injection

SQL Injection is a security flaw in web applications where attackers insert harmful SQL code through user input. This can allow them to access sensitive data, change database contents, or even take control of the system. It's important to know about SQL Injection to keep web applications secure.



Major Cybersecurity Threats & Attacks



6. Zero-Day Exploits & Advanced Persistent Threats

Zero-day exploit is a type of cyber security attack that occurs on the same day the software, hardware, or firmware flaw is detected by the manufacturer. As it's been zero days since the security flaw was last exploited, the attack is termed a zero-day exploit or zero-day attack.

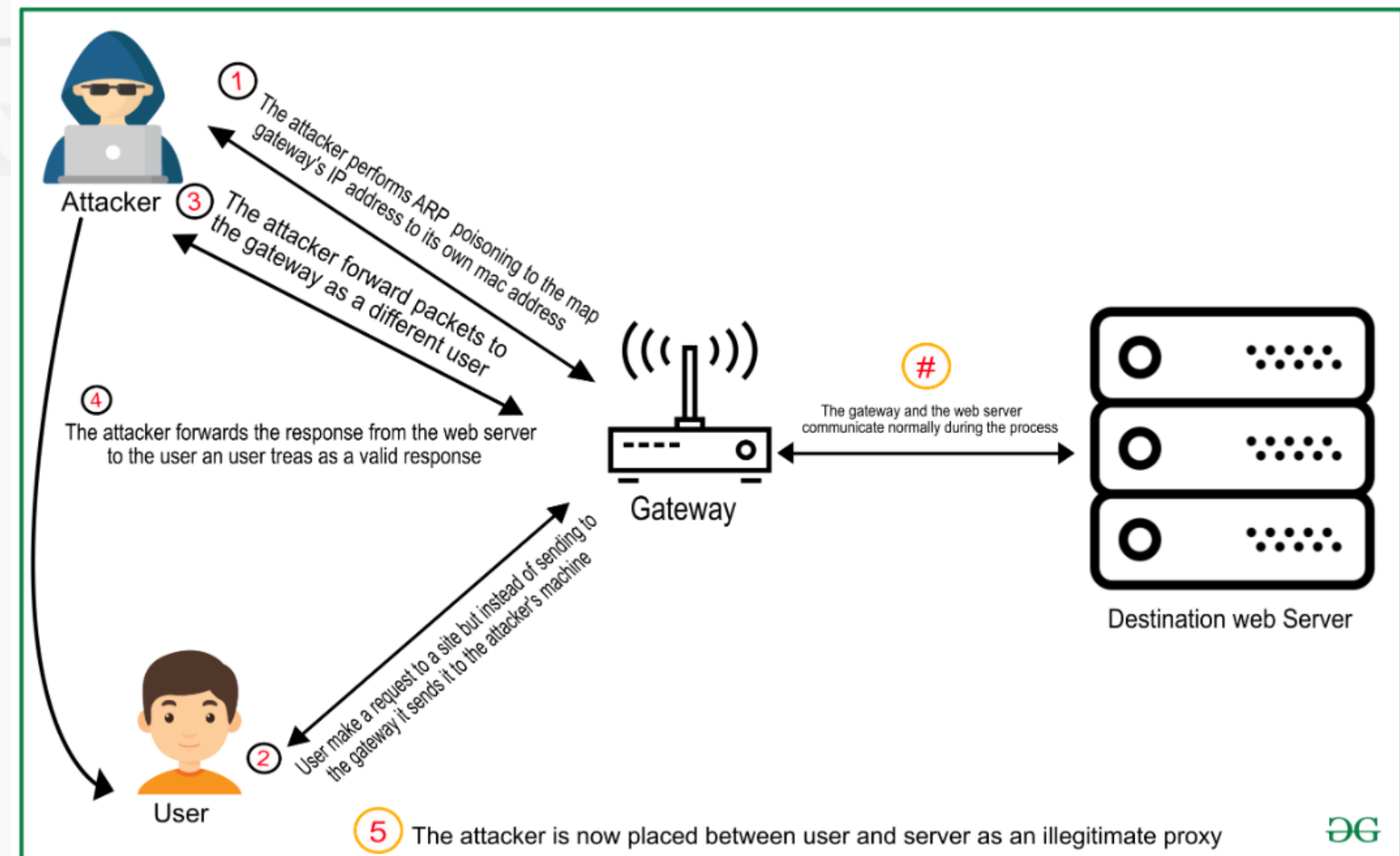
This kind of cyber-attack is considered dangerous because the developer has not had the chance to fix the flaw yet. Zero-day exploits typically target large organizations, government departments, firmware, hardware devices, IoT, users having access to valuable business data, etc.

Major Cybersecurity Threats & Attacks



7. Man-in-the-Middle (MITM) Attacks

A Man-in-the-Middle (MITM) attack occurs when an attacker secretly positions themselves between two communicating parties (e.g., your browser and a web server), so they can intercept, read, and often modify the messages passing between them all without either party knowing.



Major Cybersecurity Threats & Attacks



8. Insider Threats & Privilege Misuse

An insider threat is any threat to an organization's security that comes from within. It could be a disgruntled employee, a careless contractor or a third-party partner with too much access. Insiders already have access to sensitive systems and data, which makes them uniquely positioned to cause harm whether intentional or accidental.

Insider Attacks are carried out by people who are familiar with the computer network system and hold authorized access to all the information.

How to Stay Safe?



There are several steps you can take to protect yourself from cyber threats, including:

- **Use strong passwords:** Use unique and complex passwords for all of your accounts, and consider using a password manager to store and manage your passwords.
- **Keep your software up to date:** Keep your operating system, software applications, and security software up to date with the latest security patches and updates.
- **Enable two-factor authentication:** Enable two-factor authentication on all of your accounts to add an extra layer of security.
- **Be aware of suspicious emails:** Be cautious of unsolicited emails, particularly those that ask for personal or financial information or contain suspicious links or attachments.
- **Educate yourself:** Stay informed about the latest cybersecurity threats and best practices by reading cybersecurity blogs and attending cybersecurity training programs.